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PI Name:	Clark, Noel A. Ph.D.		
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PI Email:	noel.clark@colorado.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	303-941-1008
Organization Name:	University of Colorado		
PI Address 1:	Physics Department, CB390		
PI Address 2:	2000 Colorado Ave		
PI Web Page:			
City:	Boulder	**State:**	CO
Zip Code:	80309-0001	**Congressional District:**	2
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Contact Monitor:	Tin, Padetha	**Contact Phone:**	216-433-8164
Contact Email:	Padetha.Tin@grc.nasa.gov		
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COI Name (Institution):	Maclennan, Joseph Ph.D. (University of Colorado) Glaser, Matthew Ph.D. (University of Colorado) Stannarius, Ralf Ph.D. (Magdeburg University, Germany) Park, Cheol Soo Ph.D. (University of Colorado)		
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Task Description:

Detailed tests of theories of hydrodynamic flow, of relaxation of hydrodynamic perturbations, and of hydrodynamic interactions in 2D will be made. Freely suspended bubbles in microgravity, without islands, convection, and sedimentation represent nearly ideal, physically and chemically homogeneous 2D fluid systems for the precision study of 2D hydrodynamics. The effects of introducing islands or droplets will be studied, both as controllable inclusions that modify the flow and as markers of flow.
A second goal will be to study the behavior of collective systems of 1D layer step interfaces on 2D bubble surfaces, including the equilibrium spatial organization and interaction of islands, and the nonequilibrium coarsening dynamics of island emulsions. In addition to yielding information about a number of relatively weak physical effects (thickness-dependent surface tension and line tension, disjoining pressure, etc.), we anticipate that this will clarify the effects of dimensionality in coarsening dynamics (e.g., on dynamic scaling behavior).

We will study interactions between islands and droplets in regimes where they are very weak, in the smectic A phase and at high temperature in the smectic C phase.

The first thermocapillary experiments on homogeneous two dimensional fluids--The proposed microgravity experiments present a unique opportunity to explore thermocapillarity, the translational symmetry of the films, and the absence of convection mitigating anomalous effects to the maximum extent possible, enabling detailed studies of the thermocapillarity of 2D fluids. Study the dependence of surface tension and line tension on film thickness and Burgers vector-- Ground-based experiments indicate that equilibrium and nonequilibrium island behavior should be sensitive to this dependence, enabling critical tests of extant theoretical predictions.

We will be probing the effects of a spontaneously broken symmetry in the 2D film surface (the appearance of 2D polar, XY-like ordering and accompanying electrostatic polarization) on the interaction of islands, exploring the stability of topologically stabilized emulsions of 1D interfaces in 2D.

See also: https://

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Liquid crystals can flow like a liquid, but contain molecules that are arranged in a specific pattern, just like a crystal. Greater understanding of the physics behind these structures could lead to improved liquid crystal display devices, including devices with improved color contrast and response times. It could also advance research in high-speed electro-optic devices used to control a light beam.

HYDRODYNAMICS INTERACTION OF SMECTIC ISLANDS IN FREELY SUSPENDED SMECTIC LIQUID CRYSTAL FILMS
We study experimentally and theoretically the hydrodynamic interaction of pairs of circular inclusions in two-dimensional, uid smectic membranes suspended in air. By analyzing their Brownian motion, we find that the radial mutual mobilities of identical inclusions are independent of their size but that the angular coupling becomes strongly size-dependent when their radius exceeds a characteristic hydrodynamic length. These observations are described well for arbitrary inclusion separations by a model that generalizes the Levine/MacKintosh theory of point-force response function and uses a boundary-element approach to calculate the mobility matrix for inclusions of finite extent.

We have done measurements and theoretical modeling of the mobilities of pairs of inclusions in smectic A liquid crystal films, fluid membranes that are only a few nanometers thick. These films are very stable and provide an ideal platform for studying hydrodynamics in 2D fluids. In our experiments, we observe the Brownian motion of silicone oil droplets and smectic “islands” embedded in the films. The islands are disk-shaped, thicker regions of the film bounded by edge dislocations that can be created with diameters of between a few and several hundred μm. The oil droplets form lens-shaped rather than flat inclusions. They are essentially insoluble in liquid crystal and their sizes remain practically constant over long time intervals. This is particularly useful for making mobility measurements of very small inclusions because smectic islands less than about 10 μm in diameter tend to shrink rapidly and vanish within a few minutes, whereas silicone oil droplets of similar size remain practically unchanged for half an hour or more.

The liquid crystal used in our experiments is 8CB (4'-n-octyl-4'-cyanobiphenyl, Sigma-Aldrich), a room temperature smectic A material. The density and viscosity of 8CB are rho ~ 0.96 g/cm3 and eta = 0.052 Pa · s respectively, while the viscosity of ambient air is eta' = 1.827 × 10-5 Pa · s. Each smectic layer has a thickness of 3.17 nm. Freely-suspended films from two to six molecular layers thick, corresponding to Saffman lengths between 9 and 27 μm, were formed by spreading a small amount of the liquid crystal across a 5 mm diameter hole in a glass cover slip and were then observed using reflected light microscopy. Immediately after a film is drawn, one typically observes many islands, with a range of diameters. The film may then be gently sheared using an air jet to break larger islands into smaller ones for study. To create oil droplets, the LC film was put into a sealed chamber, where it was left until it was uniform in thickness. A double-sealed rotary pump (Welch Model 1400) was then used to reduce the chamber pressure to 3 × 10-3 Torr. After about an hour, a small amount of vaporized pump oil has made its way to the film chamber, where it eventually condenses onto the film and forms visible droplets with diameters of between 4 and 15 μm. We then restore the chamber to atmospheric pressure and observe the diffusion behavior of the droplets. Isolated pairs of islands (or oil droplets) of similar sizes and far from other inclusions and the film boundaries were selected for study.

The films are carefully leveled to minimize any gravitational drift, allowing us to record, with high spatial resolution at a typical video frame rate of 24 fps, the motion of inclusion pairs for several minutes before they diffuse out of the field of view. We use Canny’s method for edge detection and Taubin’s curve fitting algorithm to measure accurately the positions and sizes of the inclusions in each frame. The effects of any overall drift are removed analytically from the resulting trajectories. The film thickness, an integral number of smectic layers, is determined precisely by comparing the reflectivity of the film with black glass.

In order to model the effects of the long-range hydrodynamic interactions between the inclusions, we consider two circular domains of radii a and b that are subjected to external forces F1 and F2, respectively. Since we are in the low Reynolds number regime, inertial effects are unimportant and the hydrodynamics can be described using the Stokes equations. The velocity of each inclusion is a linear function of the applied forces, that of the first inclusion, for example, being given by

V1 = M11F1 +M12F2, (1)

where M11 is the self-mobility matrix and M12 the mutual mobility matrix. Since a pair of circular inclusions has mirror symmetry about the line connecting their centers, the only non-vanishing components of the mobility matrices are the diagonal elements M11rr , M11theta theta , M12rr , M12theta theta, where rr refers to the radial motion of the inclusions (along the line connecting their centers), and theta theta to tangential motion (perpendicular to this line). The mutual mobilities can be extracted from the experimental measurements by computing the cross-correlation function

delta r1(t) · delta r2(t) delta <r12(0) - s)> = 2kBT (M12rr (s) + M12theta theta(s)) t, (2)

where delta rk(t) = rk(t) - rk(0) is the displacement of the kth inclusion in time interval t, and r12(0) and s are respectively the distances between the centers of the inclusions at t = 0 and at time t.

The observation that the mutual radial mobility of two identical circular inclusions is size-independent, even at small separations, may be understood by considering the 2D velocity field around a single inclusion moving in a membrane, computed as described above for both small and large values of a/lS. For all values of a/lS, the flow fore and aft of the inclusion falls off more slowly with radial distance than the flow in the regions beside the inclusion, the precise form of the fall-off depending on the value of a/lS. The velocity gradients across the flow field in the fore and aft regions are relatively small in all cases, implying that a finite-sized second inclusion experiences the same mean flow as a point-like inclusion when the two inclusions move towards or away from each other. Furthermore, a multipole expansion of the computed velocity field shows that the contributions of the higher order multipoles to the radial mutual mobility become small when the point forces are distributed uniformly and in a mirror-symmetric way along the circumference of the (circular) inclusion, explaining why the LM response functions describe the radial mutual mobility accurately.

A similar argument holds for the tangential relative motion when the Saffman length is large but in the case of small Saffman lengths, there are significant velocity gradients in the regions directly to either side of the inclusion that result in a strong size dependence of the mutual mobility. In this case, the higher-order multipole contributions to the tangential mobility cannot be neglected, even for a symmetric distribution of point forces.

Finally, when the inclusions have different diameters, the radial mutual mobility is found experimentally to be size-dependent at all Saffman lengths. The model confirms this result and predicts similar deviations from the point-source approximation when the inclusions are not circular, for example when they are elliptical.

In summary, we have probed the hydrodynamic interactions of a pair of inclusions in a thin fluid membrane. A theory generalizing the point particle approach of Levine and MacKintosh in order to consider inclusions of finite extent and arbitrary separation reproduces the experimental mobilities obtained by measuring the Brownian motion of oil droplets and islands in thin smectic A liquid crystal films. The model confirms the surprising experimental observation that for identical circular inclusions, the mutual radial mobilities are independent of size for all Saffman lengths, while the mutual tangential mobilities depend strongly on both size and separation only when the inclusions are larger than the Saffman length.

Two-Dimensional Island Emulsions in Ultrathin, Freely-Suspended Smectic Liquid Crystal Films

We report the creation of a novel type of two-dimensional colloidal emulsion of disc-shaped domains in ultrathin freely-suspended fluid films of smectic-C liquid crystal. These stable and easily controllable emulsions are produced by repeating cycles of lateral compression and stretching of the film while maintaining shear and extensional air flow across the film. We observe a rich array of phenomena under sedimentation that includes two-dimensional phase separation, island coalescence in the condensed phase, and island and chain formation organizing into two-dimensional smectics of chains in the dispersed phase. We have measured the island size distribution, structure function, and chain spacing as a function of time. We argue that the unprecedented emulsion stability is controlled by a network of edge dislocations at the island boundaries and the elastic dipole interaction of the positive topological defects within the islands and the negative counterparts outside.

In these experiments, we are interested in the behavior of extended island emulsions and we have therefore developed a reliable method of creating large numbers of islandsat will. Nitrogen gas is directed to two hollow, stainless steel needles with 2 mm inner diameter via flexible silicone tubing with the pressures independently controlled by adjustable valves to be in the range of 0-5 cc/min. The needle pair can be translated so that air flow can be applied to any part of the film in order to create more islands from the meniscus and homogenize the island size and thickness. The needles are aligned coaxially, facing each other. The tips of the needles are 1 mm apart and positioned 2 mm above the smectic film so that the opposing air jets create shear flow in the smectic film that causes thicker domains in the film to break up into smaller ones.

The island motion is captured using either a macroview camera, which is located below the film holder, or a microview camera above. In both cases, illumination from a white light source is guided through a positive lens, reflects off a beamsplitter onto the smectic film and is reflected back through the beamsplitter to the camera. A piece of black felt put behind the beam splitter prevents unwanted reflections that would lower the contrast. An objective lens placed between the beamsplitter and the film acts both as an extra condenser lens and achives image magnification for the microview camera. The camera used for the macroview is a JVC Everio camcorder that can capture high definition videos at 1920x1080 resolution. The camera in the microview is a Watec 221S color CCD with 640x480 pixel resolution that is interfaced to a computer in order to capture video on demand for up to two days. Both cameras operate at 30 frames per second.

The film holder is enclosed to ensure that the film will be undisturbed by the air-currents in the laboratory so that the evolution of the emulsion can be studied over several hours. Although the film holder is set up to be horizontal, it possesses a small tilt, undetectable by a conventional leveling bubble yet responsible for the amazing evolution of the island emulsion in the film.

When the film is first created, it consists of domains of various thicknesses—thickest on the outer edge and thinnest in the center of the film but without any islands. The film is then sheared by extensional flow from the air jets. With the tips of the nozzles placed near the center of the film, we gradually increase the air flow, adjusting the relative strength of the two jets to achieve a four-lobe dynamical pattern. The nozzles can be moved horizontally in order to create islands and homogenize their sizes and thicknesses, to move them to the film edge, or to pull more material into the film center

After the majority of the liquid crystal at the film's edge has been pulled into the center, the spreader can be moved backward to slowly compress the film. The air needles are shifted to keep it at the center of the compressing film to keep

Task Progress:

the film homogenized, resulting in a film where most islands share similar thickness, indicated by their similar color. The average island thickness can be increased by pulling more material from the meniscus into the film with the air nozzles. Film should be kept larger than 1/3 of the size of the film holder to avoid breakage. We can then re-enlarge the film while still keeping the nozzles roughly at the center of the film. If we tune the enlarging rate to be high enough, very thin domains nucleate in the film background and the air flow from the nozzles helps combine these thin domains to be the new background which can be as thin as 2 layers (6.3 nm). Repeating this cycle of compressing-uncompressing several times results in an emulsion that fills the entire film with close to 100% islands having very similar colors and sizes in a black background, a favorable configuration for observation and analysis.

After the air from the nozzles stops flowing, the islands stop moving within about a second Sedimentation then begins with the largest, thickest islands (thickness in the order of 1 μm), sifting down through the other islands until they reach the bottom of the film holder. The velocity of the white island is about 5 μm/s and thus reach the bottom edge within 10 minutes.

The thinner islands also sediment but at a much slower rate than the faded islands because they are not as heavy and there is short range elastic repulsion between the islands. Gravity imposes a gradient on the islands distribution, making the emulsion most dense near the “white island” at the bottom edge. After some time (~10 min), the islands at the bottom, including the white islands, part of the film with no touching each other and eventually coalesce. The colored islands at the top grows slowly during this time.

Interestingly, the depleted upper region does not stay empty for long as new gray islands with thickness ranging from 10 – 100 nm pop up due to film thickening and form chains. At first, the chains are short with many branch points (dislocations), but over time, they grow in length and sometimes cover the entire width of the film. They also become smoother with longer with longer uninterrupted sub-chains between dislocations. The colored islands near the interface also form chains and right at the interface, chains can consist of both the original colored islands and the newly formed gray ones.

OBSERVATION AND ANALYSIS OF SMECTIC ISLANDS IN SPACE

Freely suspended films of smectic liquid crystals are unique examples of quasi two-dimensional fluids. Mechanically stable and with quantized thickness of the order of only a few molecular layers, smectic films are ideal systems for studying fundamental fluid physics, such as collective molecular ordering, defect and fluctuation phenomena, hydrodynamics, and nonequilibrium behavior in two dimensions (2D), including serving as models of complex biological membranes. Smectic films can be drawn across openings in planar supports resulting in thin, meniscus-bounded membranes, and can also be prepared as bubbles, either supported on an inflation tube or floating freely. The quantized layering renders smectic films uniquely useful in 2D fluid physics. The OASIS team has pursued a variety of ground-based and microgravity applications of thin liquid crystal films to fluid structure and hydrodynamic problems in 2D and quasi-2D systems. Parabolic flights and sounding rocket experiments were carried out in order to explore the shape evolution of free floating smectic bubbles, and to probe Marangoni effects in flat films. The dynamics of emulsions of smectic islands (thicker regions on thin background films) and of microdroplet inclusions in spherical films, as well as thermocapillary effects, were studied over extended periods within the OASIS (Observation and Analysis of Smectic Islands in Space) project on the International Space Station (ISS). We summarize the technical details of the OASIS hardware and give preliminary examples of key observations.

OASIS setup and technical implementation: The primary goal of the OASIS experiments on the ISS was to carry out investigations of hydrodynamic flow in smectic freely suspended films, the relaxation of hydrodynamic perturbations, and the long-term behavior of emulsions of islands and droplets. Experiments were also performed to study thermocapillary phenomena and the effects of applied electric fields and mechanical stresses on the bubbles. The microgravity conditions on the ISS prevented sedimentation and eliminated buoyancy-driven thermal convection of the surrounding air and in the liquid crystal films. A bubble diameter of 15 mm was considered sufficiently large to minimize the influence of the meniscus around the bubble-inflation needle on the dynamics of inclusions in the film. Film thicknesses were expected to be in the range between about 5 and several hundred nm (i.e., from 2 to 100 or more smectic layers), with thin films preferred as the background of island emulsions, and thicker films being more suitable for preparing droplet arrays by melting islands into the nematic or isotropic phase.We proposed to study emulsions of islands created primarily by shear flow of the films. Islands can be used as tracers to visualize flow on the bubble surface, and their size and shape can be monitored in order to characterize the dynamical behavior of 2D island emulsions. The mutual interactions of inclusions and their interactions with topological and layering defects are also of interest. We also planned to shoot picoliter-sized droplets of an immiscible isotropic liquid onto the bubbles using a microdispenser and to study their behavior. In addition to mechanical manipulations of the smectic bubbles by controlled inflation, deflation and shear flow in the film plane, the experimental apparatus was designed to allow the study of the influence of thermal gradients and electric fields. Apart from the long-term character of the experiment, the ISS provides an environment where the quality of microgravity is much better than in parabolic flights. For comparison with the parabolic flight conditions, we show a 20 s sequence of typical acceleration data measured during our experiment. The OASIS hardware was installed in the Microgravity Science Glovebox (MSG) on the ISS. The main component is the bubble chamber enclosure that contains the experimental setup, including illumination and observation elements.

PLATEAU-RAYLEIGH INSTABILITY (ISS Experiment)

We have observed what looks like the two-dimensional equivalent of the Plateau-Rayleigh instability in a thin bubble of smectic A liquid crystal (the 8CB mixture, Sample 1). In this experiment, we began by creating a thin bubble as usual. The temperature was raised locally to about 35 degrees using a heated needle located near the "north pole" of the bubble. A gentle stream of air applied across the top of the bubble (from one of the airjets) caused flow of the bubble material, which eventually entrained residual material originally located near the inflation needle, extending it into a continuous ribbon of thicker material along the flow direction that was swept around and around the bubble by the airjet. This steady-state motion of the ribbon could be continued indefinitely.

However, when the heater needle temperature was increased to 45 degrees, a dramatic transition in the shape of the liquid ribbon occurred, with it breaking up into a chain of small disks (the two-dimensional equivalent of droplets) which then continue to flow around the film. It is proposed that this increase in temperature lowers the line tension (the two-dimensional equivalent of surface tension) of the ribbon sufficiently that it becomes unstable to small shape perturbations and pinches off into islands, much the same way that a water jet behaves in three dimensions.

ISLAND EMULSION COARSENING (ISS Experiment)

Three-dimensional (3D) coarsening dynamics in emulsions, foams, and other non-equilibrium systems is an important and relatively well-studied problem. Bubbles of smectic liquid crystal, investigated in the OASIS (Observation and Analysis of Smectic Islands in Space) experiments on the International Space Station, offer a well-characterized, homogeneous platform for the study of both equilibrium and non-equilibrium behavior of collective systems of 1D interfaces in 2D. There have been very few studies of such phenomena in fluids.

OASIS investigators at the University of Colorado have already made the first observations of the coarsening of island emulsions on the two-dimensional surface of bubbles of smectic liquid crystal created on ISS. Air jets are used to create extensional flow in an initially homogeneously thin bubble film. This draws excess material from the bubble meniscus (which is confined to a small region near the needle used to blow the bubble) onto the bubble surface. The air flow also breaks this thicker material up into small, disk-like inclusions called islands that then form a 2D emulsion on the bubble surface.

On Earth, such island emulsions would rapidly sediment to the lowest point of the bubble and coalesce but in microgravity, coarsening of the emulsion can be observed free of gravitational effects. Early observations on smectic A liquid crystal bubbles 15 mm in diameter confirm that an emulsion of small islands evolves to a significantly coarser ensemble of islands within about one hour. Much of the coarsening appears to be the result of island coalescence events. Further investigations will be carried out to determine how much Ostwald ripening might also be occurring and to characterize the dependence of coarsening on dimensionality.

The OASIS experiment, which was launched on SpaceX-CRS-6 in April 2015, is being carried out in collaboration with investigators from the University of Magdeburg in Germany, the Russian Academy of Sciences in Moscow, and project scientists from NASA's Glenn Research Center.

OSTWALD RIPENING AND COLLAPSE OF ISLANDS IN SMECTIC BUBBLE (ISS Experiment)

The coarsening dynamics of one-dimensional (1D) interfaces in two-dimensional (2D) systems is of considerable theoretical interest, but detailed laboratory studies have been hindered by a paucity of appropriate and convenient experimental systems. We have recently made experimental observations of 2D coarsening in which thickness islands grow via coalescence and Ostwald ripening in molecularly thin, tethered smectic liquid crystal bubbles. We report here on the dynamics of island growth and collapse without coalescence in island emulsions at locations far from the meniscus around the bubble inflation needle. This Ostwald ripening of essentially 2D smectic A and C islands occurs as a result of the system’s tendency to reduce its energy through a decrease in the total length of the boundaries of the islands, a process that is generally dependent on the (thickness-dependent) surface and line tension, the disjoining pressure of the island, and the excess air pressure inside the bubble. Smaller islands generally have a higher disjoining pressure than their larger neighbors, which results in permeative flow from the smaller to the larger islands. We will describe our observations of Ostwald ripening behavior in smectic bubbles in microgravity and compare these with the dynamics of island coarsening dynamics in flat smectic films.In smectic bubbles pressure in the films is mainly determined not by the meniscus, but by the excess pressure in the bubble. This induces a change in the physics of collapse and ripening of islands in smectic films.

TIME AND SPACE PERIODIC STRUCTURES OF ISOTROPIC INCLUSIONS (DROPLETS) ON SMECTIC BUBBLES (ISS Experiment)

Collective behavior of isotropic droplets dispersed over a spherical smectic bubble was studied in microgravity conditions. Droplets can form two-dimensional hexagonal structures. We observe spatial and temporal periodicity of such structures of droplets, and investigate their spatial arrangement. We found a peculiar periodic in time ordering of the droplets, related to one-dimensional motion of droplets with nonuniform velocity. This behavior can be considered as a classical analogue of discrete time crystal.

In this work we study effects related to self-organization of particles (droplets of isotropic liquid) leading to formation of hexagonal structures. The investigations are performed in microgravity conditions on the American segment of the International Space Station (ISS) in the framework of OASIS project. Various types of collective droplet behavior were found. Positional and orientational ordering of droplets in the hexagonal structure is analyzed. The widths of radial and azimuthal droplet position, orientation distribution functions increase with distance. Under velocity gradient in the film we observed periodic appearance and disappearance of hexagonal ordering of droplets. This behavior can be considered as an analogue of “time crystals” observed in periodically driven non-equilibrium systems.

In the experiment Smectic A (SmA) liquid crystal MX12160 with direct phase transition from the SmA to the isotropic (Iso) phase (SmA – 51.1ºC – Iso) was used. The first stage of the experiment was the creation of a smectic bubble of spherical shape. A small amount of the material in the SmA phase was deposited at the end of a thin metallic needle. Then excessive pressure was created inside the needle, which allowed blowing a smectic bubble. Applying different pressure bubbles of different size could be obtained. In present experiments bubbles with diameter about 1.5 cm were used. Two video cameras were employed to capture the images of the whole bubble and a micro image of the bubble region which was used to register structures formed by droplets. The bubble had regions of uniform thickness (they have different brightness), on which registration of micro images was made.

On heating close to the temperature of the SmA-Iso transition, droplets of isotropic phase are born in the film. In experiments on board ISS in microgravity conditions (~10-3 g) the droplets remained in the film and did not sediment to the bottom of the bubble as it happens in terrestrial conditions. Droplet size and their density can differ in different regions of the bubble. Registration of micro images of groups of droplets was performed with droplets about 15 μm in size. In some regions of the film where the surface density of droplets was greater or about 5·102 mm-2, a quasi hexagonal structure of droplets could be observed. The structure resembles the hexagonal ordering of droplets observed earlier in flat free-standing smectic films in terrestrial conditions. The distances between droplets in the hexagonal structure can be sufficiently larger than the droplet size. The nature of long-range interactions leading to its formation up to now is not clear (note that in SmA films there are no in-plane interactions related to the c-director, which stabilize ordered structures of inclusions in films of tilted smectics). It could be capillary forces, interaction due to smectic elasticity, Van der Waals interaction, or critical fluctuations of the smectic order parameter relevant in the vicinity of the phase transition point. We analyze the ordering of droplets in the hexagonal structure. Positions of the droplets in the first three coordination circles are plotted with respect to the origin. Measurements were performed on droplets. Distances to nearest, next nearest and next-next nearest neighbors in the quasihexagonal structure were measured. It should be noted that in the experiment the micro images were taken from a non-central part of the bubble which is

oriented obliquely with respect to the focus plane of the camera. This induced some distortion in the image, namely shrinkage of the distance in the direction parallel to the inclination plane. We are confident that after this procedure the distortion due to tilt of the film plane with respect to the camera is corrected.

At motion of the droplets we observed behavior which can be considered an analogue of the “time crystal.” The concept of time crystals was put forward by Frank Wilczek. According to his works, this unique state of matter can exist both in quantum and classical case. The possibility to form a time crystal is closely connected with the fundamental concept of spontaneous symmetry breaking used in different areas of physics, ranging from conventional condensed matter phase transitions to cosmological Kibble mechanism of topological defect formation. The essential feature of a time crystal is that it restores the same state at specific moments of time. This idea as a generalization of the concept of symmetry breaking was a challenge for theory and experiment. Later it was shown that discrete time crystals can exist only in non-equilibrium systems. Recently, discrete time crystals were realized in several quantum systems, but as far as we aware our result is the first observation of time crystalline behavior in classical systems.

In our case close to the temperature of the SmA-Iso transition regions of different thickness can transform and change their shape. As a result, the material of the film and the droplets also moves. In the case of a uniform drift of the droplets in the film plane the hexagonal structure is conserved. Interesting effects were observed when the velocity of the droplets was nonuniform. The structure formed by droplets in which the hexagonal order periodically disappears and reappears can be considered as an analogue of the classical “time crystal.” In our observation nearly constant gradient of mean velocity was created only in nearest three rows of droplets. It is worth noting that if the constant gradient of velocity were created in a larger number of rows, one can expect a much richer behavior, e.g., appearance of a line of hexagons perpendicular to the motion of droplets in discrete moments of time (which are determined by the velocity of the row of hexagons with the smallest velocity). A process similar to the described above can be observed in shear melting of three-dimensional colloidal crystals, where a so-called intermediate two-dimensional “sliding layer structure” is found. In summary, two-dimensional structures of droplets in smectic films were observed in microgravity conditions. Stationary hexagonal arrangement of droplets is observed and analyzed quantitatively. When the droplets in the film move with a gradient of velocity, periodic appearance and disappearance of the hexagonal structure was found. Microgravity creates favorable conditions for experiments. In the normal gravity conditions a number of difficulties should be overcome to realize a similar experiment even in flat free-standing film geometry. The experiment on spherical smectic bubbles in normal gravity is hardly feasible at all since sedimentation of droplets occurs. The observed behavior exhibits a number of main features of time crystals. The structures are time dependent, appear at certain moments of time in a non-equilibrium dynamical system. The question still exists about the reason of stability of the hexagonal structure in absence of in-plane orientational interactions and about the possibility to observe structures of droplets with different type of ordering.

Bibliography Type:	Description: (Last Updated: 12/04/2024)
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